UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which statement immediately exits a loop?
   A. cin
   B. return 0
   C. continue
   D. break

2. What is the index of the first element of a C++ array?
   A. -1
   B. 1
   C. The array size
   D. 0

3. How many times does a while loop execute when its condition is false initially?
   A. Until the user stops it
   B. One
   C. Zero
   D. Two

4. Which structure chooses between two alternative paths?
   A. if...else
   B. struct
   C. array
   D. for

5. Which declaration creates a floating-point variable named price?
   A. char price;
   B. int price;
   C. bool price;
   D. float price;

6. Which values can a bool variable hold?
   A. any decimal
   B. any character
   C. any integer
   D. true or false

7. Which term describes a finite sequence of precise steps that solves a problem?
   A. Algorithm
   B. Array
   C. Compiler
   D. Variable

8. Which item is NOT a programming paradigm?
   A. Procedural programming
   B. Object-oriented programming
   C. Debugging
   D. Structured programming

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9. Which identifier is valid in C++?
   A. 2total
   B. int
   C. total score
   D. total_score

10. Which input pattern stops when the user enters -1?
   A. A function prototype
   B. A sentinel-controlled loop
   C. A fixed for loop
   D. A switch statement

PART B (60 MARKS)

QUESTION 1

This question tests programming concepts and error terminology.

a) Define compiler and interpreter.

(3 marks)

b) State THREE differences between a compiler and an interpreter.

(2 marks)

QUESTION 2

A program reads three integer values and displays the largest value.

a) Draw a flowchart to find the largest of three values.

(6 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

(7 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(5 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
   s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(5 marks)

QUESTION 5

A class records ten student marks in an array.

a) Declare an integer array of ten marks and read the marks from the user.

(6 marks)

b) Write a program segment to calculate and display the average, highest mark and number of marks above the average.

(10 marks)

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PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A catering company charges RM12.50 per meal. Orders of 50-99 meals cost RM11.50 each, 100-199 cost RM10.50 each, and 200 or more cost RM9.50 each.

Write a complete C++ program that reads the customer name and quantity, validates the quantity, calculates the total, displays a receipt, and repeats until the user chooses to stop. At the end, display the total quantity and total sales.

(15 marks)

END OF QUESTION PAPER